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Supervisory Control and Data Acquisition (SCADA)

Project Report

Title:
Scada-Based Smart Conveyor Industrial Sorting System

Group: CEEE222

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1.0 INTRODUCTION

1.1 Executive Summary

In most factories and distribution centers, the parcel sorting process is one of the important processes before the parcel is sent to the designated destination. However, this process can still involve manual inspection by workers, especially to determine the weight or size of the parcel. This method is time-consuming, requires a lot of manpower and can cause errors during the sorting process. Therefore, the SCADA-Based Smart Conveyor Industrial Sorting System project was developed to help automate the parcel sorting process so that the sorting work can be done faster, more accurately and systematically.

This project uses a PLC as the main controller that receives signals from four sensors, namely Proximity Sensor, Load Cell Weight Sensor, Ultrasonic Height Sensor and IR Photoelectric Sensor. Based on the sensor readings, the PLC will process the data using a ladder diagram to control the conveyor motor, pneumatic pusher and indicator lights. Parcels weighing more than 800 g will be sorted into Bin 1, while parcels with a height of more than 50 cm will be sorted into Bin 2. Parcels that do not exceed both limits will continue their journey to Bin 3.

In addition, this project also uses SCADA Human Machine Interface (HMI) to allow operators to monitor the system status in real time. The HMI display provides functions such as Start and Stop control, sensor status, timer, counter, trend graph for Height Sensor and alarm when the parcel exceeds the set weight limit. Through this project, the concept used can provide an overview of the application of PLC and SCADA in industrial automation systems and how this technology can help improve the efficiency of the sorting process and reduce the need for manual parcel inspection.

1.2 Problem Statement

Despite the increasing development of automation technology, there are still several critical constraints in the management of parcel flow in the industry. Among them is the failure of multi-parameter classification, where most conventional sorting systems are only able to separate goods based on a single criterion, such as color, while modern industrial needs require more detailed classification, including height, weight, and material type, such as metal or non-metal. Without a combination of integrated digital and analog sensors, the sorting process becomes less accurate and is very prone to misclassification of goods, which is detrimental to the company. Next is the irregularity of mechanical motion control, which has difficulty in controlling the movement of mechanical components such as pneumatic actuators and servo motors in a systematic and coordinated manner. Without the integration of logic programming systems that use delay timers and counters, the movement of these components often becomes inefficient, causing goods to collide or not reach the sorting location at the right time. In addition, the lack of data transparency and real-time monitoring, industries that still rely on manual inspection face great problems in monitoring the status of operations and obtaining accurate object weight data directly. The absence of a SCADA HMI interface equipped with automatic data logging functionality makes the data recording process blurred and has a high risk of human error in warehouse inventory management.

1.3 Objective

- To develop a system that can automatically sort goods by weight and height using 4 types of sensors.
- To create a ladder diagram for a SCADA module using a timer and counter to move the pneumatic pusher and conveyor motor in an orderly manner.
- To evaluate the performance by designing a SCADA HMI screen display to monitor real-time operations, view graphs of items or parcels weights, and store report data.

1.4 Scope Project

The purpose of this project is to design and implement SCADA-Based Smart Conveyor Industrial Sorting System for automatic parcel sorting and real time monitoring. The system is constructed using a conveyor belt coupled with sensors, pneumatic actuators and a SCADA interface to carry out the sorting operation automatically. The project implements four primary sensors, namely, proximity sensor, load cell weight sensor, ultrasonic height sensor, and IR photoelectric sensor. These sensors are utilized for detection of items, measurement of the weight and height, and counting the number of parcels successfully passing through the conveyor. The acquired data are processed by the control system and displayed on the SCADA interface for monitoring.

The output devices are the functioning of the conveyor motor, two pneumatic pushers and indication lights. The pneumatic pushers are used to redirect items to different collecting bins according to the sorting criteria. The indicator lights indicate the operation state of the system. The SCADA interface is also designed to provide real-time monitoring, display of system status and recording of operational data throughout the sorting process. The project is confined to sorting of items based on weight and height only and is evaluated under a controlled laboratory setting utilizing prototype scale equipment. Other parcel attributes such as colour, shape or material type are out of reach for this project.

2.0 PROPOSED SOLUTION

2.1 Block Diagram

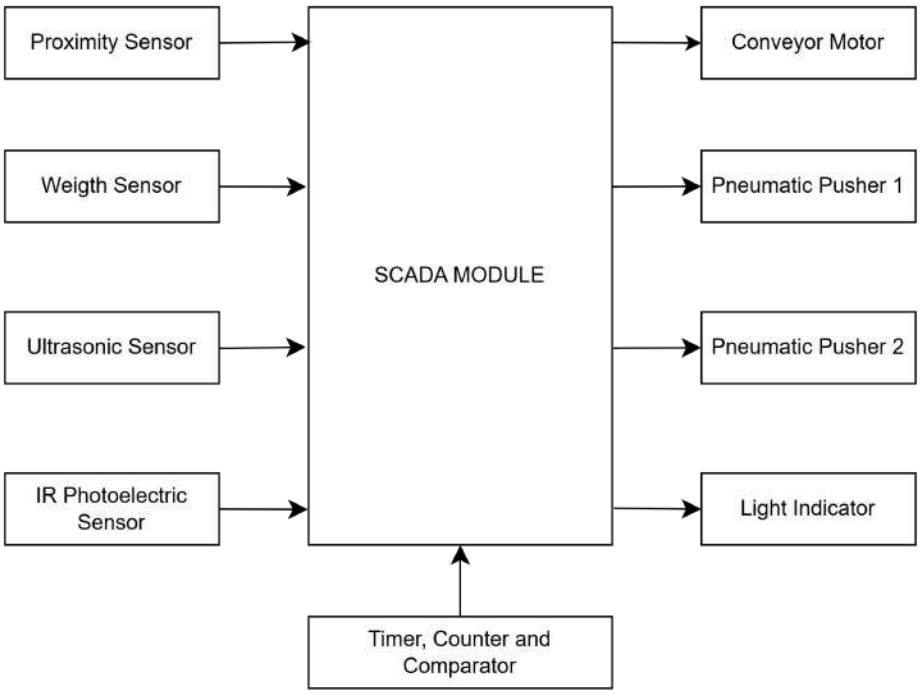


Figure 2.1: Block Diagram of the system

Figure 2.1 shows the block diagram of the SCADA-Based Smart Conveyor Industrial Sorting System. The system consists of four input components namely proximity sensor, weight sensor, ultrasonic sensor and IR photoelectric sensor. All the input then will send the data signal directly to the SCADA Module. The function of proximity sensors is to detect the presence of a parcel on the conveyor, while the weight sensor and ultrasonic sensor provide the necessary information for the sorting process by measuring the physical characteristics of the parcel. The IR photoelectric sensor is positioned at the end of the conveyor to detect and count the parcels that successfully pass through the sorting process. In addition, the timer, counter, and comparator are incorporated into the system to support the control operation by providing timing functions, counting the number of parcels, and comparing the sensor readings with the predefined conditions.

The SCADA module serves as the central unit of the system by receiving input signals from all sensors and processing them according to the programmed control logic. The SCADA

module will send control signals to the output devices which are the conveyor motor, Pneumatic Pusher 1, Pneumatic Pusher 2, and the light indicator. The conveyor motor is used for moving parcels on the conveyor system, while Pneumatic Pusher 1 and Pneumatic Pusher 2 act as a mechanical pusher to sort the parcel by separating it to their designated destinations. The light indicator provides a visual indication of the system operating status, allowing operators to easily identify whether the system is running or stopped. In general, the system operates by detecting the incoming parcel, processing the sensor information, performing the required sorting action, and monitoring the movement of parcels throughout the conveyor. At the same time, the SCADA module updates the system status and records the operational data, allowing the overall sorting process to be monitored and controlled efficiently.

2.2 Flowchart

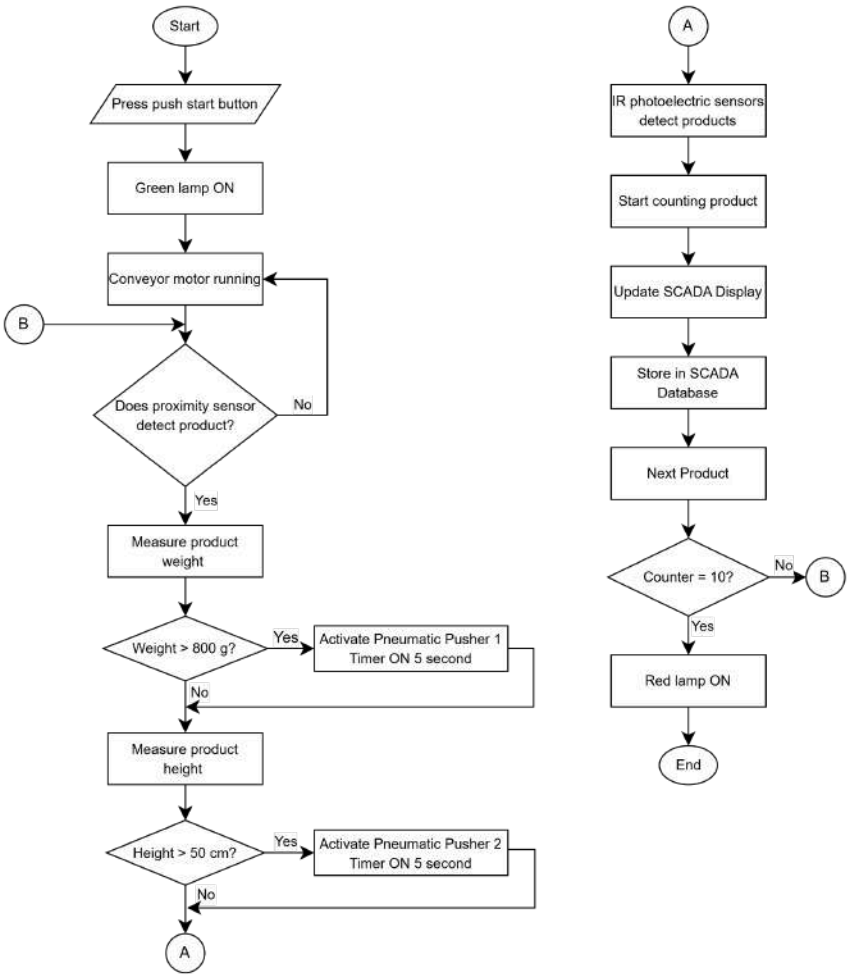


Figure 2.2: Flowchart of the system

Figure above shows the overall system workflow of the project. The system begins when operator press the push start button. Then a green lamp is turned ON indicate that the system has started running. Then the conveyor motor is activated and the conveyor belt starts to move. Once the object enter the conveyor line, the proximity sensor start to act and detect the presence of the object. After the object is detected, the weight sensor measures the weight of the parcel. The value then compared with the value that has been set. If the parcel weight is more than 800g, Pneumatic Pusher 1 is activated to push the parcel into the first bin along with a timer that is on for 5 second. If the parcel does not meet the weight condition, it will continue moving along the conveyor to the next inspection stage.

Next, the ultrasonic sensor measures the height of the parcel. The measured height is compared with the preset value. If the parcel height is more than 50cm the Pneumatic Pusher 2 is activated to push the parcel into the second bin along with a timer that is on for 5 second. However, if the parcel does not satisfy both the weight and height conditions, it will continue moving along the conveyor until it reaches the IR photoelectric sensor. The IR photoelectric sensor detects every parcel that passes through the sorting process and starts counting the total number of parcels. At the same time, the counting value is updated on the SCADA display and stored in the SCADA database for monitoring purposes. After that, the system waits for the next parcel and repeats the same process continuously. Once the parcel counter reaches the preset value of 10, the system automatically stops and the red indicator lamp turns ON to indicate that the sorting process has been completed.

2.3 Wiring Diagram

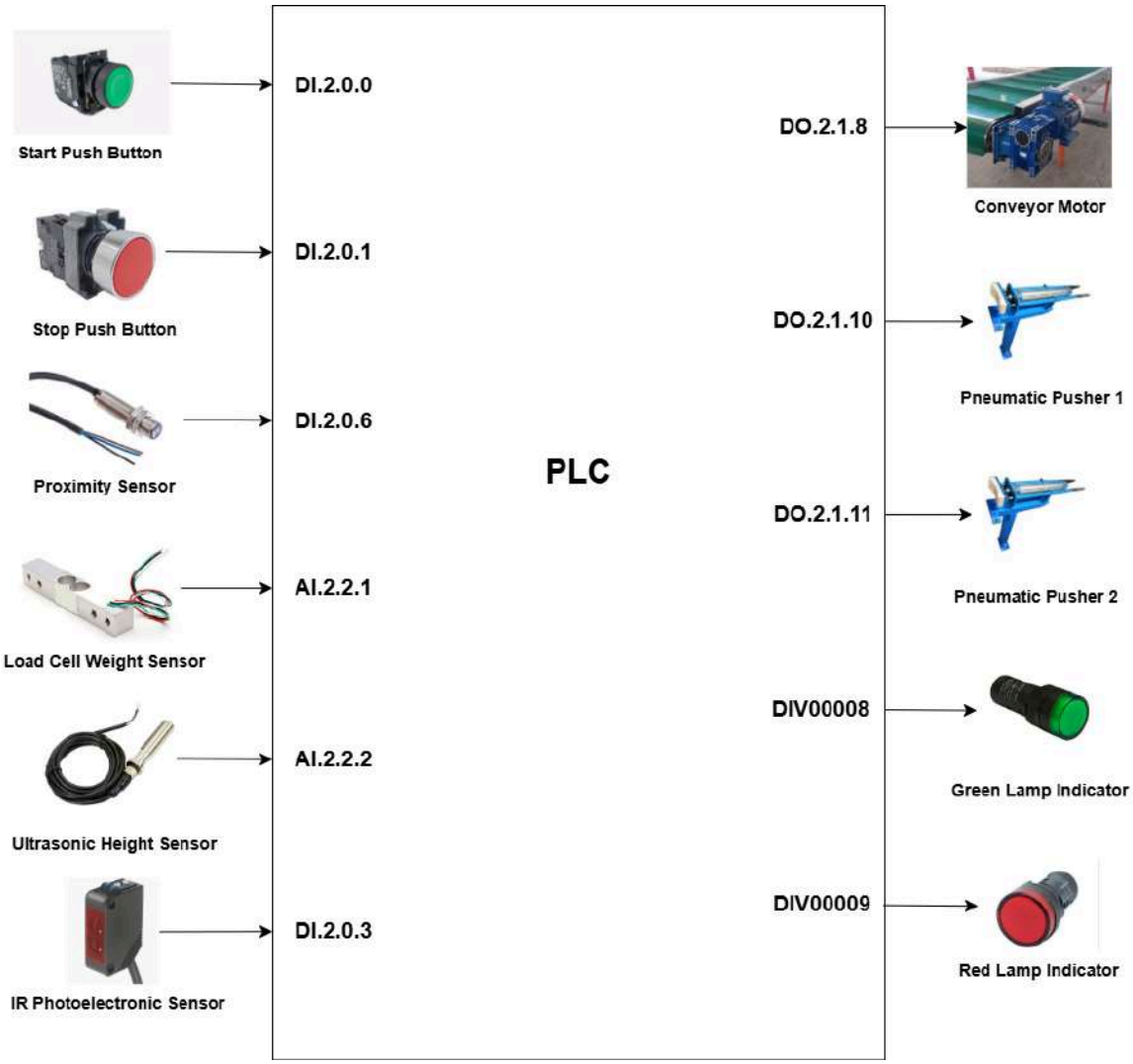


Figure 2.3: Wiring diagram of the SCADA-Based Smart Conveyor Industrial Sorting System.

Figure 2.3 shows the wiring diagram for the SCADA-Based Smart Conveyor Industrial Sorting System. This wiring diagram explains the connection between input and output devices with the PLC as the main controller of the system. All signals from the sensors will be sent to the PLC to be processed according to the ladder diagram program before the PLC controls each output device based on the specified conditions.

On the input side, the system uses six devices, namely Start Push Button, Stop Push Button, Proximity Sensor, Load Cell Weight Sensor, Ultrasonic Height Sensor, and IR Photoelectric Sensor. The Start Push Button is used to turn on the system, while the Stop Push

Button functions to stop the entire system operation. The Proximity Sensor detects the presence of parcels at the beginning of the conveyor before the measurement process is carried out. The Load Cell Weight Sensor and Ultrasonic Height Sensor are connected as analog inputs because both sensors produce reading values that vary according to the weight and height of the parcel. Next, the IR Photoelectric Sensor is connected as a digital input to count the number of parcels that have passed the end of the conveyor.

On the output side, the PLC controls the Conveyor Motor, Pneumatic Pusher 1, Pneumatic Pusher 2, Green Lamp Indicator, and Red Lamp Indicator. The Conveyor Motor functions to move the parcels along the conveyor. Pneumatic Pusher 1 will be activated when the weight of the parcel exceeds the set value, while Pneumatic Pusher 2 will be activated when the height of the parcel exceeds the specified limit. Green Lamp Indicator shows that the system is in operation, while Red Lamp Indicator shows that the system is stopped or inactive.

In addition, each device is connected to a different PLC input and output address as shown in the diagram. The use of this address makes it easier for the PLC to identify each device when executing the ladder program. With a systematic wiring arrangement, all components can communicate with the PLC more orderly, thus ensuring that the process of parcel detection, separation and monitoring can be carried out automatically.

2.4 Sensor and Actuators

I. Proximity Sensor



Figure 2.4.1: Proximity Sensor

The proximity sensor is used as the first detection sensor in the system. Its main function is to detect the presence of a parcel when it enters the conveyor belt. Once a parcel is detected, the sensor sends a signal to the SCADA module to initiate the sorting process. This ensures that the parcel inspection only starts when an object is present on the conveyor. In addition, the sensor helps improve the accuracy of the system by preventing unnecessary operation when there is no parcel on the conveyor. It also allows the conveyor system to operate continuously and efficiently throughout the sorting process.

II. Weight Sensor

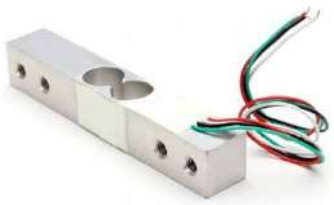


Figure 2.4.2: Load Cell Weight Sensor

The implementation of the load cell weight sensor allows for the precise measurement of parcel mass as items traverse the conveyor belt. Data obtained from this sensor is relayed to the SCADA module, facilitating a comparison against established parameters to determine the sorting logic for Pneumatic Pusher 1 or subsequent processing. This component ensures high measurement integrity, which is vital for the operational accuracy of the automated classification sequence. Furthermore, the weight values are visualized on the SCADA HMI, enabling real-time monitoring and data recording of the inventory flow.

III. Ultrasonic Sensor



Figure 2.4.3: Ultrasonic sensor

Ultrasonic sensors are used to measure the height of packages after the weight inspection process. It detects the distance between the sensor and the package to determine the height of the package. The measured value is then sent to the PLC for comparison with the preset height before a sorting decision is made. By using this sensor, packages with different heights can be identified without making physical contact with the object. This increases the reliability of the measurement while reducing the possibility of damage to the package during inspection. In industrial applications, the DFRobot URM14 (SKU: SEN0358) as shown in Figure 2.4.3 is one of the most commonly used ultrasonic sensors. This sensor has an ultrasonic frequency of 200 kHz. This specification is suitable for measuring the height of objects in automatic sorting systems.

IV. IR Photoelectric Sensor



Figure 2.4.4: IR sensor

The IR photoelectric sensor is installed at the end of the conveyor to detect every parcel that successfully passes through the sorting process. Its main function is to count the number of parcels and send the counting information to the SCADA module. The recorded data is displayed on the SCADA interface and stored for monitoring purposes. This sensor ensures that every

parcel passing through the conveyor is counted accurately without manual intervention. The collected data can also be used by the operator to monitor the overall parcelion progress in real time.

V. Conveyor Motor



Figure 2.4.5: Conveyor motor

The conveyor motor is the main actuator that drives the conveyor belt throughout the sorting process. It transports the parcels from one inspection station to another in a continuous manner. The motor operates according to the control signals received from the SCADA module and stops automatically when the preset parcel count is achieved or when the system is turned off. A stable conveyor movement is important to ensure that each sensor can inspect the parcel accurately at the correct position. This contributes to a smooth and efficient sorting operation.

VI. Pneumatic Pusher



Figure 2.4.6: Pneumatic pusher

The figure above shows the example of a pneumatic pusher that is used in the industry. The pneumatic pusher is used as the sorting actuator in the system. It receives a control signal

from the SCADA module and pushes the parcel into the designated bin whenever the sorting conditions are satisfied. Two pneumatic pushers are used in this project, where Pneumatic Pusher 1 is responsible for the first sorting stage, while Pneumatic Pusher 2 performs the second sorting stage. Both pushers are activated only after the timer condition is fulfilled to ensure the parcel is in the correct position before being pushed. This helps improve the accuracy and consistency of the sorting process

VII. Light Indicator



Figure 2.4.7: Light Indicator

The light indicator provides a visual indication of the system operating status. The green light indicates that the system is running normally, while the red light indicates that the system has stopped after completing the sorting process or when the system is switched off. This allows the operator to monitor the system status easily without accessing the SCADA interface. The indicator also improves operational safety by providing a clear visual signal of the current system condition. Light indicators are commonly used in industrial automation systems to indicate different machine operating conditions such as running, standby, warning, or fault status. However, in this project, only two indicator lights are implemented, where the green light indicates that the system is operating normally and the red light indicates that the system has stopped after completing the sorting process or when the system is switched off. As a result, operators can quickly identify whether the system is active or inactive from a distance.

2.5 3D Design

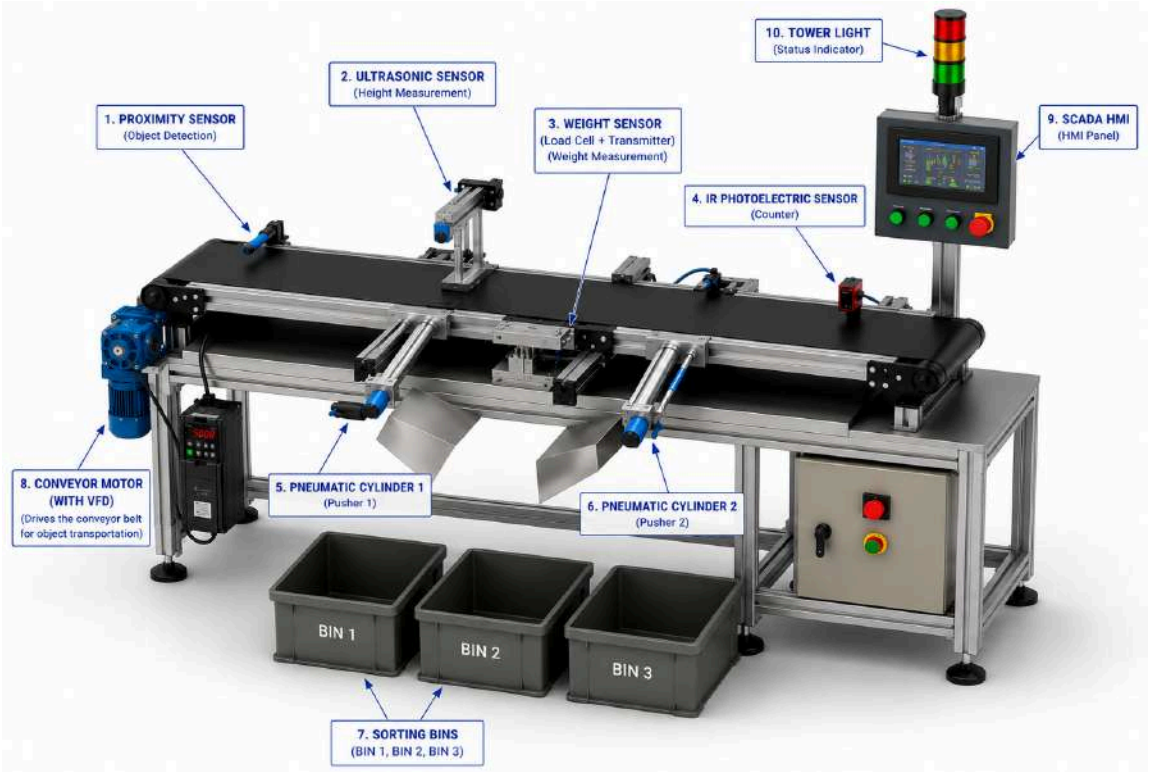


Figure 2.5.1: A 3D design of the system

Figure 2.5.1 illustrates the 3D design of the SCADA-Based Smart Conveyor Industrial Sorting System. The design presents the overall arrangement of the conveyor structure, sensors, actuators, sorting bins, and SCADA monitoring station. The system is developed to perform automatic sorting operations based on object height and weight while providing real-time monitoring through the SCADA interface. The detailed functions of each component used in the system are described below.

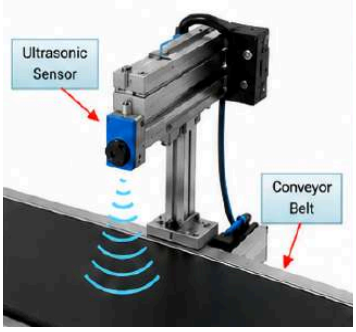


Figure 2.5.2: Ultrasonic Sensor

The Ultrasonic Sensor is used to measure the height of the object moving on the conveyor belt. The sensor emits ultrasonic waves and calculates the distance based on the reflected signal received from the object surface. The measured height data is sent to the SCADA controller for classification purposes.

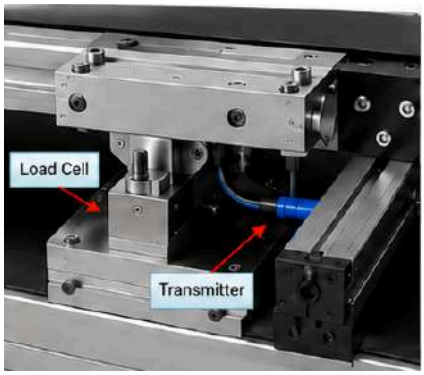


Figure 2.5.3: Weight Sensor

The figure above shows the weight sensor which is responsible for measuring the weight of the object. The load cell converts the applied force into an electrical signal, while the transmitter converts the signal into an analog value that can be processed by the SCADA system. This measurement is used to determine whether the object exceeds the predefined weight limit.

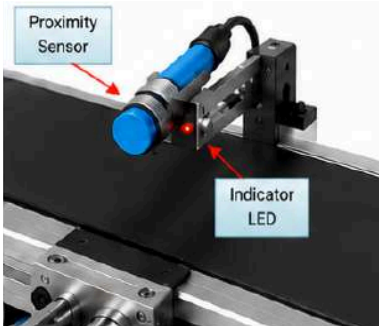


Figure 2.5.4: Proximity Sensor

The Proximity Sensor detects the presence of an object entering the conveyor system. Once an object is detected, the sensor sends a signal to activate the sorting sequence and initiate data acquisition from the other sensors.

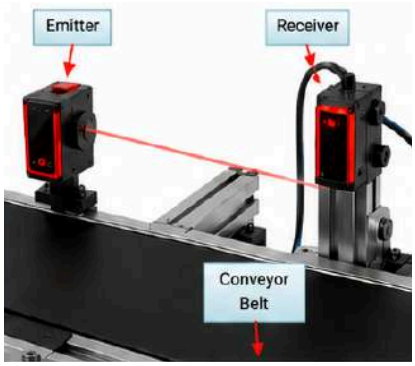


Figure 2.5.5: IR Photoelectric Sensor

The IR Photoelectric Sensor is installed at the end of the conveyor line to count the number of objects that have successfully passed through the sorting process. The counting data is recorded for inventory tracking and reporting purposes.

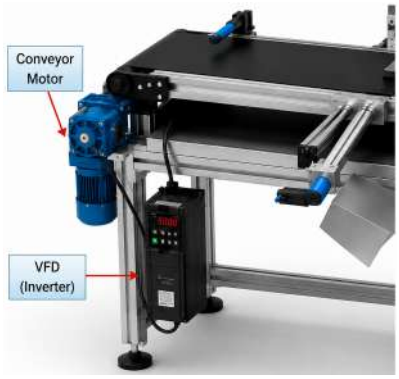


Figure 2.5.6: Conveyor motor

The conveyor motor drives the conveyor belt, allowing objects to move through the inspection and sorting stations. The Variable Frequency Drive (VFD) controls the motor speed, providing smooth operation and adjustable conveyor movement.

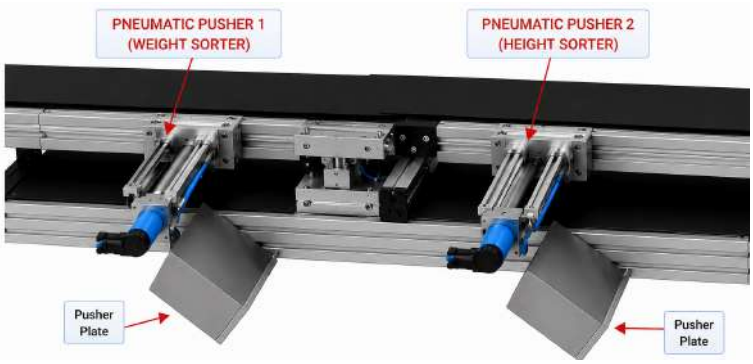


Figure 2.5.7: Pneumatic pusher 1 & 2

The pneumatic pushers automatically divert objects from the conveyor into designated collection bins based on the inspection results. Pneumatic Pusher 1 performs weight-based sorting, while Pneumatic Pusher 2 performs height-based sorting, ensuring that objects are classified according to the predefined weight and height requirements.



Figure 2.5.8: Alarm or tower light

Figure above shows the alarm which functions to display the operating condition of the system through different coloured lights.As mentioned above in the sensor and actuator section in Figure 2.4.7, only two indicator lights are implemented in this project simulation, where the green light indicates that the system is operating normally and the red light indicates that the system has stopped after completing the sorting process or when the system is switched off. But if implemented in the real industry, it is more recommended and appropriate to use a light tower like Figure 2.5.8. It helps operators quickly identify normal operation, warnings, or fault conditions.

3.0 RESULT AND DISCUSSION

3.1 Assignment List

	Name	Value	Address	Type	Comment	Initi...	Mo...
	Add a tag						
1	Start_PB		DI.2.0.0	Bool	Digital Input 0		512
2	Stop_PB		DI.2.0.1	Bool	Digital Input 1		513
3	IR_Sensor		DI.2.0.3	Bool	Digital Input 3		515
4	Prox_Sensor		DI.2.0.6	Bool	Digital Input 6		518
5	Weight_Sens...		AI.2.2.1	Float	Analog Input 4..20mA 1		578
6	Height_Sensor		AI.2.2.2	Float	Analog Input 4..20mA 2		580
7	Conveyer_M...		DO.2.1.8	Bool	Digital Output 8		552
8	Pneumatic_1		DO.2.1.10	Bool	Digital Output 10		554
9	Pneumatic_2		DO.2.1.11	Bool	Digital Output 11		555
10	M1		DIV00000	Bool			20480
11	Status_T1		DIV00001	Bool			20481
12	Preset_T1		AIV00001	Long			20482
13	Value_T1		AIV00000	Long			20480
14	Status_T2		DIV00002	Bool			20482
15	Value_T2		AIV00002	Long			20484
16	Preset_T2		AIV00003	Long			20486
17	Status_C1		DIV00003	Bool			20483
18	Value_C1		AIV00004	Long			20488
19	Preset_C1		AIV00005	Long			20490
20	M2		DIV00004	Bool			20484
21	M3		DIV00005	Bool			20485
22	HMI_StartPB		DIV00006	Bool			20486
23	HMI_StopPB		DIV00007	Bool			20487
24	Green_Light		DIV00008	Bool			20488
25	Red_Light		DIV00009	Bool			20489
26	Weight_Bit		AIV00006	Int			20492
27	Height_Bit		AIV00007	Int			20494
28	Weight_Scal...		AIV00008	Int			20496
29	Height_Scaled		AIV00009	Int			20498

Figure 3.1: Assignment list of Twinsoft tags used for the ladder diagram and SCADA HMI.

Figure 3.1 shows the assignment list or tag list that has been built into the Twinsoft software. This list is used to connect each input, output and internal data device with a specific address so that all tags can be used in ladder diagrams and SCADA HMI. Each tag has a name, address, data type and comment to facilitate the development process and program inspection.

On the input side, tags such as Start_PB, Stop_PB, IR_Sensor and Prox_Sensor use the Boolean data type because the device only has two states, namely ON or OFF. For example, Start_PB will signal ON when the start button is pressed, while Prox_Sensor will be active when a parcel is detected on the conveyor. The weight sensor and height sensor use analog addresses, namely Weight_Sensor and Height_Sensor, because both sensors produce reading values that change according to the weight and height of the parcel.

On the output side, the Conveyor_Motor, Pneumatic_1 and Pneumatic_2 tags are used to control the movement of the conveyor and the parcel separation process. Conveyor_Motor will be active while the system is running, while Pneumatic_1 and Pneumatic_2 will be activated based on the results of the weight and height comparison. The Green_Light and Red_Light tags are also used to indicate the system status, whether the system is operating or has stopped.

In addition to the physical input and output tags, this assignment list also has several internal tags such as M1, M2 and M3. These memory tags are used in ladder diagrams to store temporary status or control the sequence of system operations. Tags such as Status_T1, Preset_T1, Value_T1, Status_T2, Preset_T2 and Value_T2 are used to monitor the operation of Timer 1 and Timer 2. For the counting function, the Status_C1, Value_C1 and Preset_C1 tags are used to show the status, current value and set value of the counter.

The HMI_StartPB and HMI_StopPB tags allow the system to be controlled via the SCADA HMI display without relying solely on physical push buttons. In addition, the Weight_Bit and Height_Bit tags store raw values from analog inputs, while the Weight_Scaled and Height_Scaled are used to store values after the scaling process. This process is important so that the bit values from the sensor can be converted to real units that are easier to understand, such as grams and centimeters.

Overall, this assignment list helps ensure that each system component has a clear address and function. The organized arrangement of tags also facilitates the process of writing ladder diagrams, monitoring via SCADA HMI and the troubleshooting process if errors occur during simulation.

3.2 Ladder Diagram



Figure 3.2.1: Start, Stop and Conveyor Motor Control Ladder Diagram

Figure 3.2.1 shows the ladder diagram for the main control of the system, which is to start and stop the entire conveyor operation. The system can be started via the physical Start Push Button or the HMI Start Button on the SCADA display. When one of these buttons is pressed, the memory bit M1 will be activated. The M1 bit is used as a latching function so that the system remains in the ON state even if the Start button has been released.

To stop the system, the operator can use the physical Stop Push Button or the HMI Stop Button. When one of the Stop buttons is pressed, the logic flow to M1 will be interrupted and M1 will be OFF. In addition, a status counter is also included in the main rung to ensure that the system can be stopped automatically when the number of parcels reaches a preset value. When M1 is ON, the Green Light and Conveyor Motor will be activated. The Green Light indicates that the system is running, while the conveyor motor moves the parcels through the inspection and separation area. When M1 is OFF, the Green Light and conveyor motor will stop, while the Red Light will light to indicate that the system is stopped or inactive.

The use of the latching function helps ensure more stable system operation because the operator does not have to press the Start button continuously. In addition, the provision of physical controls and control via HMI gives the operator more options to operate the system. This ladder section is important because all other processes such as sensor reading, scaling, sorting and counting can only be carried out when M1 is in the ON state.

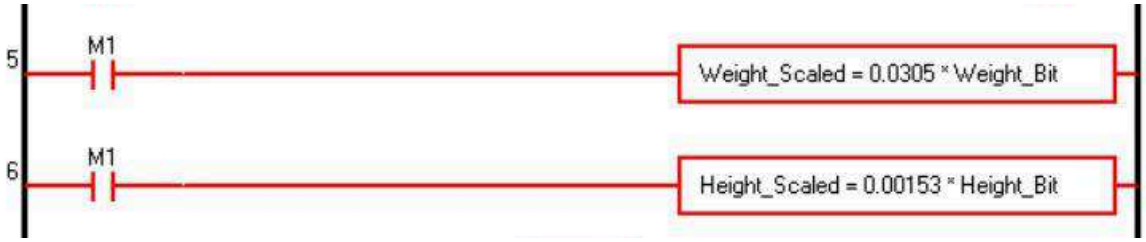


Figure 3.2.2: Analog Scaling Ladder Diagram for Weight and Height Sensors

Figure 3.2.2 shows the analog scaling process for Weight Sensor and Height Sensor. Both sensors produce readings in the form of analog values or bit values that are received by the PLC. These bit values cannot be used directly as weight values in grams or height values in centimeters. In this project, the PLC analog input is assumed to use 16-bit resolution. Therefore, a scaling formula is used to convert the raw sensor readings into values that are easier to understand and use in the program. The formula allows the bit value from the analog input to be converted to the actual weight value used in the comparison process.

This scaling process is only carried out when the system is in the ON state via bit M1. The value that has gone through the scaling process is then used in the comparator to determine whether the weight of the parcel exceeds 800g or the height of the parcel exceeds 50cm.For more details on calculating the values placed in the scaling ladder as in Figure 3.2.2. In Figure 3.2.3 the calculation has been made and assumes using 16 bits and range for weight and height. The use of scaling is very important because PLCs originally only accept data in the form of digital numbers.

$y = mx + c$

Actual Value = m(Bit) $+ c$, resolution = $2^{16} = 65536$

Bit Range : 0 to 65535

① Weight Scaling

Weight Range : 0 to 2000g

$0 = 0m + c$

$c = 0$ #

$2000 = 65535\,m + 0$

$m = 2000/65535$

$m = 0.0305$

Weight = 0.0305 (Bit)

② Height Scaling

Height Range : 0 to 100 cm

$0 = 0m + c$

$c = 0$ #

$100 = 65535\,m + 0$

$m = 100/65535$

$m = 0.00153$

Height = 0.00153 (Bit)

Figure 3.2.3: The details of Calculation for Height and Weight Scaling Ladder

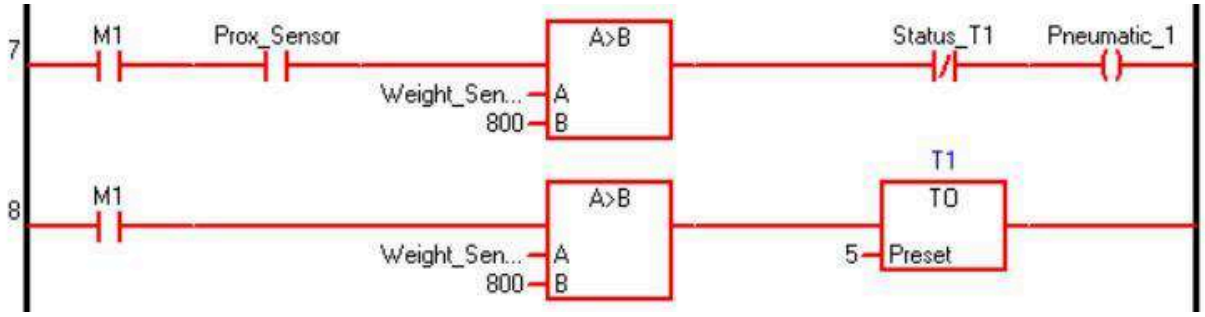


Figure 3.2.4: Weight Sorting Ladder Diagram Using Comparator and Timer

Figure 3.2.4 shows the ladder diagram for the weight-based parcel sorting process. This process starts when the system is ON and the Proximity Sensor detects the presence of a parcel on the conveyor. The Proximity Sensor is used to ensure that the comparison process is only performed when there is a parcel in the inspection area.

Next, the reading from the Weight Sensor is compared with a preset value of 800g using the comparator function A > B. In this comparison, the value A represents the parcel weight reading, while the value B is the set limit value. If the parcel weight exceeds 800g, the comparator condition will be true.When the condition is met, Timer T1 will start counting for 5 seconds. The timer is used to give the parcel time to move from the sensor area to the position of Pneumatic Pusher 1. After the timer expires, the status of T1 will change and Pneumatic Pusher 1 will be activated to push the heavy parcel to Bin 1.

The use of the comparator ensures that only parcels that exceed the weight limit will be selected for the sorting process. Meanwhile, the timer helps coordinate the position of the parcel with the pneumatic movement so that the pusher is not activated too early or too late. With a combination of Proximity Sensor, comparator, timer and pneumatic, the weight separation process can be carried out automatically and more systematically without requiring an operator to manually check the weight of the parcel.

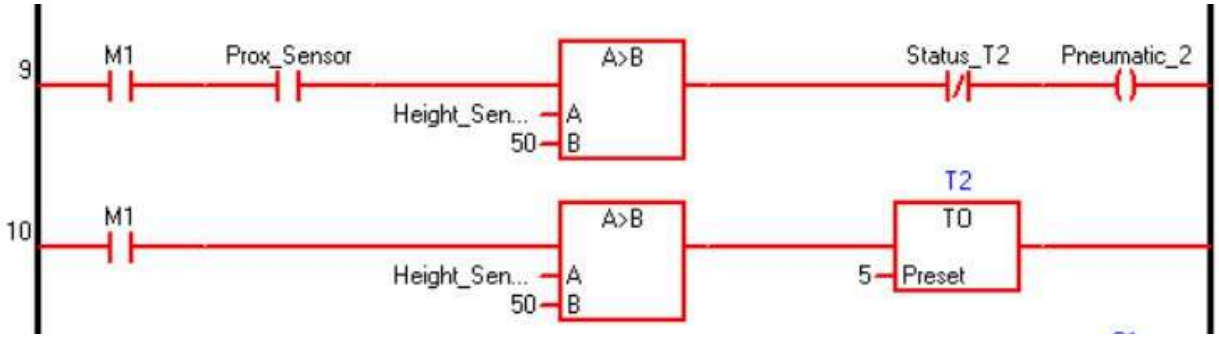


Figure 3.2.5: Height Sorting Ladder Diagram Using Comparator and Timer

Figure 3.2.5 shows the ladder diagram for the height based parcel sorting process. Similar to the weight sorting process, this operation only occurs when the system is ON and the Proximity Sensor detects the presence of a parcel on the conveyor. The reading from the Height Sensor is compared to a preset value of 50cm using the comparator A > B. Value A is the parcel height reading, while value B is the height limit that has been set. If the parcel height exceeds 50cm, the comparator condition will be true.

After that, Timer T2 will start counting for 5 seconds. The purpose of this timer is to give the parcel time to move to the Pneumatic Pusher 2 position. When the timer expires, Pneumatic Pusher 2 will be activated and push the parcel that is too high to Bin 2.

If the parcel height does not exceed 50cm, Pneumatic Pusher 2 will not be activated and the parcel will continue to move on the conveyor. parcels that do not exceed the weight and height limits will continue their journey to the normal path or Bin 3. The combination of Height Sensor, comparator and timer helps ensure that only parcels that exceed the height limit will be moved. Using a timer also helps ensure that the rejection process occurs at the appropriate time.

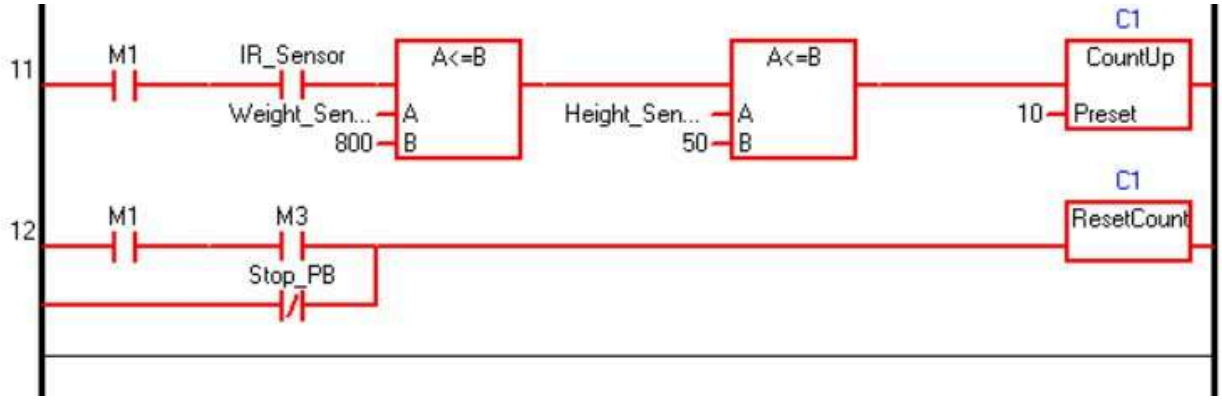


Figure 3.2.6: parcel Counting Ladder Diagram

Figure 3.2.6 shows the ladder diagram for the process of counting the number of parcels passing through the main conveyor path. The counting process uses an IR Photoelectric Sensor installed near the end of the conveyor. Each time a parcel passes the sensor, the IR Sensor will send a signal to the PLC. Before Counter C1 adds a count, the program will check two conditions using a comparator $A \leq B$. The first condition is that the weight of the parcel must not exceed 800 g, while the second condition is that the height of the parcel must not exceed 50 cm. This means that only parcels that are not categorized as too heavy or too high will be counted by Counter C1.

When all conditions are met and the IR Sensor detects the parcel, Counter C1 will add a count. The preset counter value is set to 10, which means the system will count up to ten parcels per operation cycle. The current counter value can be displayed on the SCADA HMI to facilitate the operator in monitoring the number of parcels that have been processed in real time. This ladder also has a Reset Count function. The counter can be reset when M3 is activated or when the Stop Push Button is used according to the logic that has been set. After the reset is performed, the value of Counter C1 will return to zero and the system can start a new counting cycle.

The use of counters helps in the process of monitoring the number of parcels and can be used for batch control, parcelion records and inventory. In addition, the connection of the counter with the HMI allows the operator to know the number of parcels that have passed through the system without having to do manual counting.

3.3 WebForm Studio (SCADA HMI)

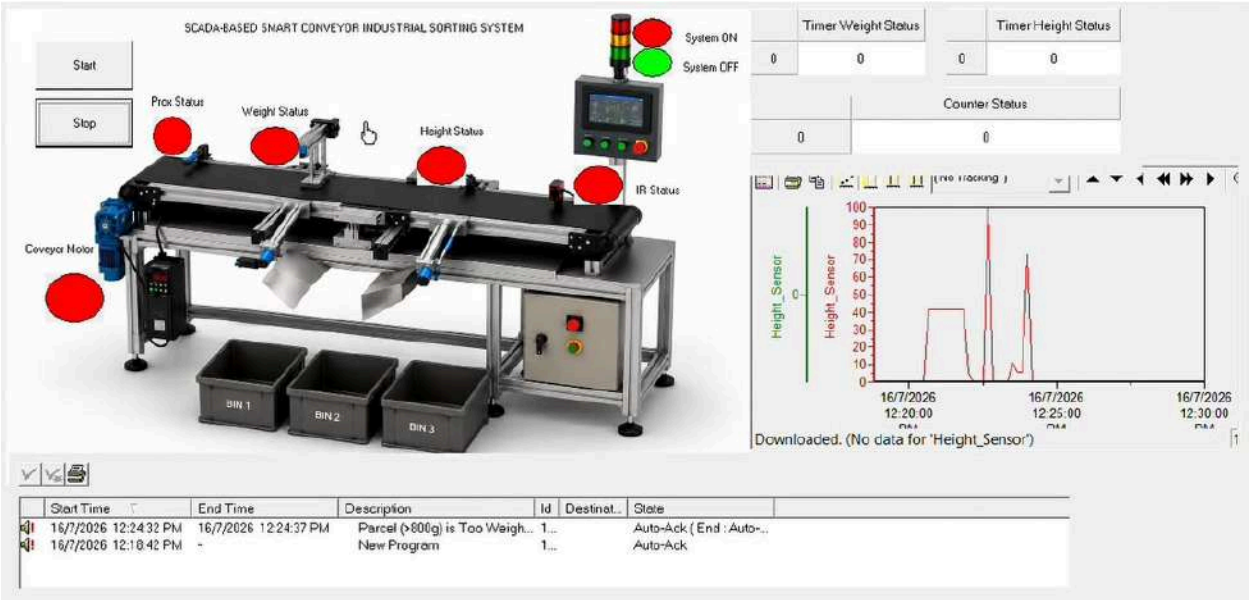


Figure 3.3.1: Initial Condition of the SCADA HMI

Figure 3.3.1 shows the initial state of the SCADA HMI display before the system is started. In this state, the conveyor motor, proximity sensor, weight sensor, height sensor and IR sensor are in an inactive state. The red light in the status section indicates that the device is still OFF, while the Timer Weight, Timer Height and Counter Status values are at zero.

Even though the system has not been started, the trend graph and alarm list still display records that have been saved from previous operations. The display is historical data in SCADA and does not indicate that the sensor is active at that time. This function allows the operator to review changes in sensor readings and alarm events that have occurred even if the system has been stopped. The system will only start operating after the Start button is pressed via the HMI display or physical push button. When need a new graph or real time data graph after press the start push button, only need to right click on the HMI display and press download (If the graph does not change).

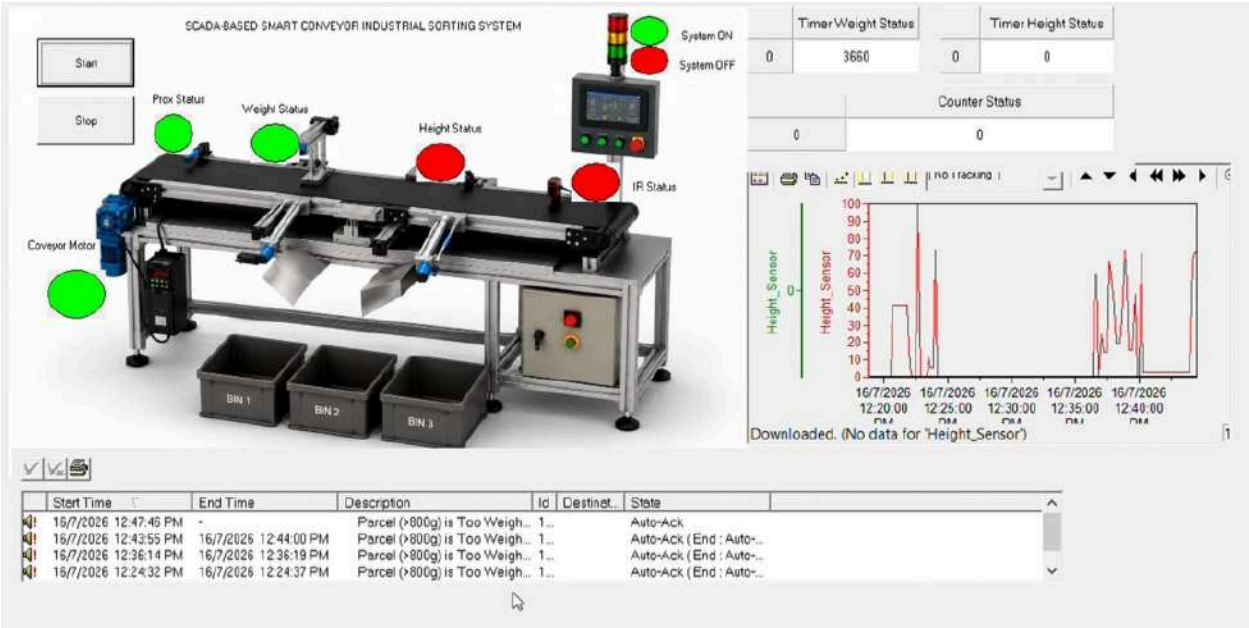


Figure 3.3.2: SCADA HMI During Weight Sorting Operation with Alarm Messages.

Figure 3.3.2 shows the HMI status while the sorting process based on the weight of the parcel exceeding the set load is in progress. The System ON light and the conveyor motor indicator are green, indicating that the system and conveyor are operating. The Proximity Status is also green because the parcel has been detected in the inspection area.

The Weight Status indicator is green when the parcel weight reading exceeds the preset value, which is 800 g. When this occurs, the PLC will activate the comparator logic and start the Timer Weight. The current value of the timer can be seen on the Timer Weight Status display. After the set time period expires, Pneumatic Pusher 1 will be used to push the parcel to Bin 1.

At the same time, an alarm message related to the parcel exceeding 800 g is recorded in the alarm log section. The record contains the start time, end time and event description, which is "Parcel (>800g) is Too Weight!!!". This display shows that SCADA not only functions to control the system, but also helps the operator identify and record the condition of the parcel that is too heavy.

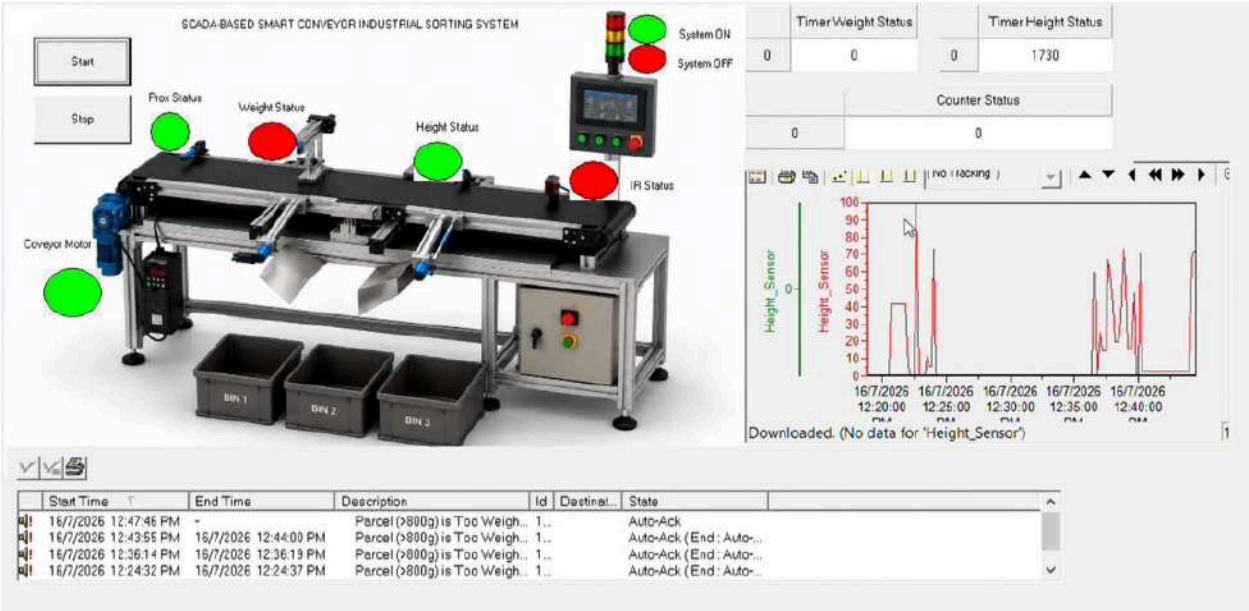


Figure 3.3.3: SCADA HMI During Height Sorting Operation with Height Sensor Trend Graph.

Figure 3.3.3 shows the SCADA HMI display during the parcel separation process based on the set height. The System ON, conveyor motor and Proximity Status lights are green, indicating that the system is operating and the parcel has been detected on the conveyor. In this situation, the Height Status changes to green because the reading from the Height Sensor exceeds the preset value, which is 50 cm. The PLC will compare the reading using the comparator function before activating the Timer Height. The current value of the timer is displayed in the Timer Height Status section to allow the operator to monitor the time calculation process before Pneumatic Pusher 2 is activated to push the parcel to Bin 2.

In addition, the trend graph on the right side of the HMI shows the Height Sensor readings recorded in real time throughout the system operation. Each change in the height value will be displayed in the form of a graph over time, allowing the operator to see the pattern of changes in the sensor readings during the sorting process. This function facilitates the process of monitoring sensor performance and helps identify if there are abnormal readings or disturbances in the sensor. This recorded data can be used as a reference during the system analysis process and future maintenance work.

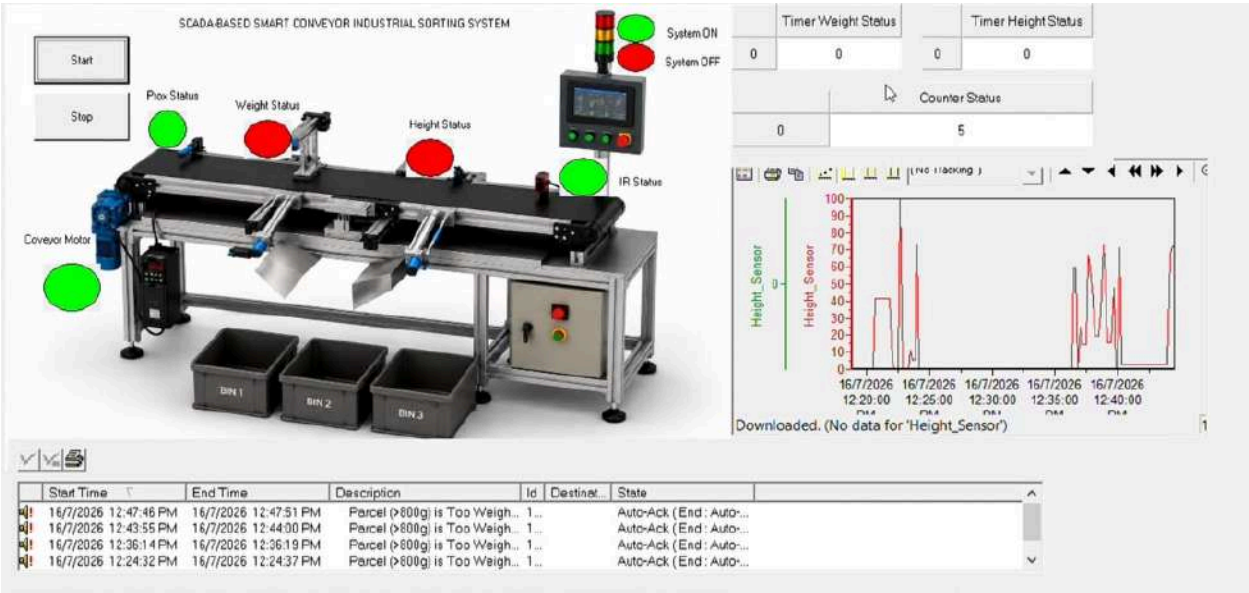


Figure 3.3.4: parcel Counting and IR Sensor Monitoring on the SCADA HMI.

Figure 3.3.4 shows the process of counting parcels using the IR Photoelectric Sensor. The system and conveyor are still operating, as indicated by the System ON and Conveyor Motor indicators which are green. When the parcel reaches the end of the conveyor, the IR Photoelectric Sensor will detect the parcel and the IR Status indicator will change to green. The signal from this sensor is sent to Counter C1 in the ladder diagram to increment the count value by one. In Figure 3.3.4, the Counter Status shows a value of 5, which means that five parcels have been successfully counted in the current operation cycle.

The Weight Status and Height Status are in red because the parcel does not exceed the weight limit of 800 g or the height limit of 50 cm. Therefore, the parcel is not rejected by the pneumatic and continues to move through the main path to the bin. The real-time counter display makes it easy for the operator to monitor the number of parcels without having to do manual counting.

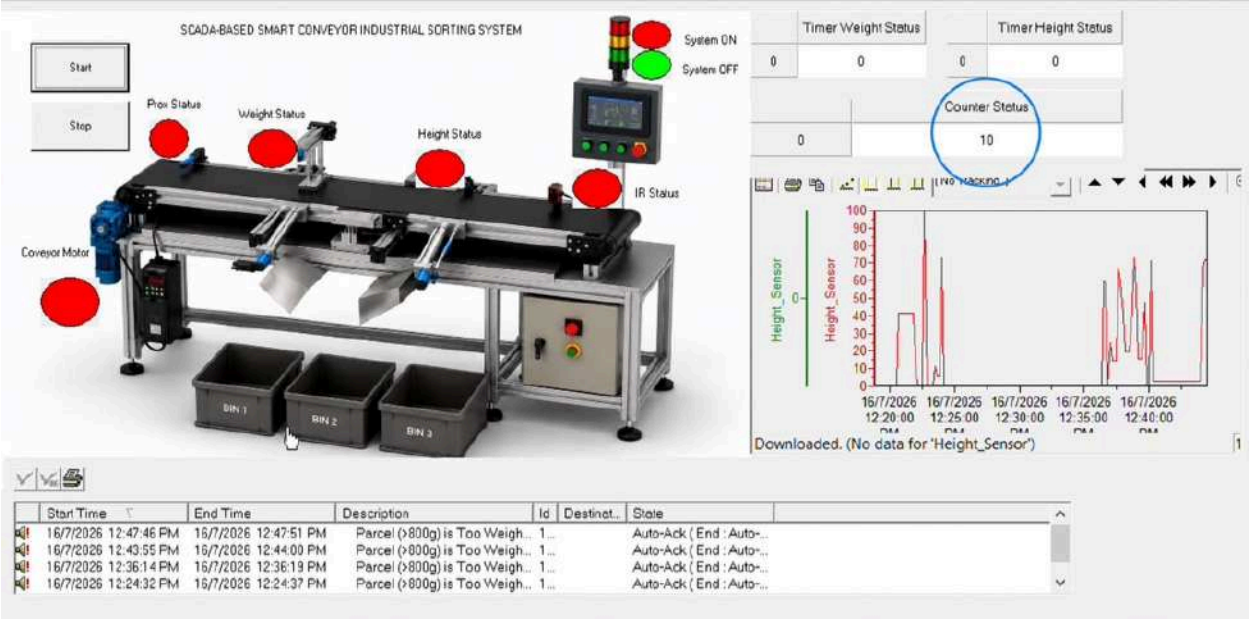


Figure 3.3.5: Automatic System Stop When the Counter Reaches the Preset Value.

Figure 3.3.5 shows the system state when the Counter Status has reached the preset value, which is 10. When the number of parcels reaches ten units, Status C1 in the ladder diagram will change and disconnect the latching of the memory bit M1. This condition causes the conveyor motor and the entire sorting process to stop automatically. In the display, the conveyor motor, proximity sensor, weight sensor, height sensor and IR sensor indicators are in red, indicating that all processes have been stopped. Timer Weight and Timer Height also return to zero values because there is no more sorting process in progress.

This automatic stop function is used as a batch control to limit the number of parcels to ten units per cycle. In this project simulation, the preset counter value is set to 10 units assuming 100 units of parcel for actual operations in the industry. After the counter is reset to zero, the system can be restarted for a new cycle. This function helps control the parcelion volume and prevent the system from continuing to operate after the counting target has been reached.

4.0 CONCLUSION

4.1 Safety and Environmental Consideration

Safety is one of the important considerations in the development of the SCADA-Based Smart Conveyor Industrial Sorting System. Throughout the operation, the conveyor motor, pneumatic pushers, and sensors are controlled automatically by the SCADA system to reduce direct human interaction with moving mechanical parts. This helps minimize the risk of workplace accidents and improves operator safety during the sorting process. In addition, the light indicator provides a clear visual indication of the system status, allowing operators to easily identify whether the system is running or stopped before performing any maintenance or inspection. Proper wiring, secure mounting of electrical components, and regular inspection of the sensors and actuators are also important to ensure safe and reliable system operation. Operators should also follow the standard operating procedures (SOP) when handling the system to prevent accidental injuries during operation or maintenance. Furthermore, routine maintenance of the electrical and mechanical components is recommended to ensure that the system continues to operate safely and efficiently over time.

From an environmental perspective, the implementation of an automated sorting system helps improve resource utilization by reducing sorting errors and minimizing parcel wastage. The use of sensors and automatic control also allows the sorting process to be carried out more efficiently, reducing unnecessary conveyor operation and energy consumption. Furthermore, the SCADA system records operational data, enabling operators to monitor system performance and identify any abnormal conditions that may affect energy efficiency. These features contribute to a more efficient and sustainable industrial operation while supporting better resource management. In addition, accurate parcel sorting helps reduce unnecessary material disposal caused by sorting errors, which can contribute to waste reduction. The implementation of automation technology also supports sustainable manufacturing practices by improving process efficiency while reducing the consumption of energy and resources.

4.2 Innovation and Future Improvement

The proposed SCADA-Based Smart Conveyor Industrial Sorting System demonstrates the integration of industrial automation technologies to improve the parcel sorting process. The project is capable of performing automatic parcel detection, sorting, counting, and real-time monitoring with minimal human intervention by integrating sensors, pneumatic actuators, and a SCADA monitoring system. The proposed system enhances operational efficiency, minimizes human errors, and offers a more systematic sorting process in comparison to traditional manual sorting methods. The implementation of the SCADA system also enables operators to monitor the system status and parcel count through a single interface, making the operation more convenient and efficient.

Additional sensing technologies, such color sensors or machine vision cameras, can be added to the system in the future to increase its ability to categorize objects based on color, form, or surface problems. Internet of Things (IoT) technology can also be integrated to enable remote monitoring and control of the system through mobile devices or cloud platforms. Other than that, a database reporting feature can be added to generate daily or monthly parcelion reports automatically, allowing better parcelion analysis and maintenance planning. These improvements would increase the flexibility, intelligence, and overall performance of the proposed system in real industrial applications.

4.3 Limitation

Several limitations were identified during the development and testing of this project. The system is designed to sort parcels based only on weight and height, making it unsuitable for applications that require sorting based on other characteristics such as colour, shape, or material type. In addition, the accuracy of the sorting process depends on the performance and calibration of the sensors. Any misalignment or inaccurate sensor readings may affect the overall sorting performance. Regular sensor calibration and proper installation are therefore important to maintain consistent sorting accuracy throughout the operation. In addition, routine maintenance should also be carried out to ensure that all sensors continue to operate properly over an extended period of use.

Furthermore, the project was developed and tested under laboratory conditions using a limited number of sample parcels. Therefore, the system has not been evaluated for continuous operation or under actual industrial environments where factors such as higher conveyor speed, vibration, dust, and varying parcel conditions may influence the system performance. Further testing and system enhancement are recommended to improve the reliability and applicability of the system for industrial use. This would help ensure that the system can perform consistently and meet the operational requirements of real industrial applications. Testing the system with a wider range of parcel types and operating conditions would also provide a better evaluation of its overall performance and reliability.

4.4 Conclusion

Overall, the SCADA-Based Smart Conveyor Industrial Sorting System project was developed as an automation application that uses PLC and SCADA to sort parcels based on weight and height. This system uses several sensors as input to the PLC before controlling the conveyor and pneumatic pusher through a ladder program that has been built. The SCADA HMI display is used to monitor system status, sensor readings, counters, timers, trend graphs and alarm messages in real time. This project provides exposure to the use of automation technology in the sorting process and helps understand how PLC, SCADA, sensors and actuators work together to produce a more organized and systematic control system.

5.0 REFERENCES

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